25900703 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCAQC202 - ANATOMY AND PHYSIOLOGY OF FINFISHES AND SHELLFISHES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Give an example of skin of fishes performing respiratory function. [K] / [CO1]
2. How do mature ovaries appear in breeding season? [U] / [CO3]
3. What is Mermaid's purse of elasmobranch? [C] / [CO5]
4. Name the two portal systems in fish? [K] / [CO1, CO2]
5. Where are the testes located in shrimp? [U] / [CO3]
6. What is the primary function of the esophagus in finfishes? [U] / [CO1]
7. Which morphometric paramater in fishes is most commonly used for accurate growth analysis? [K] / [CO5]
8. Name one method to study spawning season. [C] / [CO5]
9. What are blood sinuses in the bivalve circulatory system? [U] / [CO1, CO2]
10. In crabs at which maturity stage is the vas deferens occupy the entire body cavity? [U] / [CO3]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

11.	Distinguish between Potamodromous and Oceanodromous migration of fishes?	[An] / [CO1]
12.	Discuss the excretion of nitrogenous waste in finfishes.	[U] / [CO3]
13.	Explain the different types of social behavior in fishes with suitable examples.	[U] / [CO1]
14.	Describe the structure of the heart of teleosts, with a suitable diagram?	[U] / [CO1, CO2]
15.	Write a note on the natural food of bivalves. Describe the feeding mechanism in bivalves	[K] / [CO4]
16.	Write a note on swim bladder in teleost fishes.	[U] / [CO1]
17.	Draw the structure of fish eye and label any 4 parts	[An] / [CO3]
18.	Describe the role of excretion in maintaining ionic balance in both shrimps and bivalves.	[K] / [CO3]
19.	Explain the mechanism of respiration in Teleosts?	[U] / [CO1, CO2]

[6x5 = 30]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

20.	Describe the role of the stomach, gastric mill, hepatopancreas, midgut, and hindgut in the digestion and absorption of food in shrimp.	[U] / [CO1]
21.	Describe the structure and function of major sense organs (eye, ear, lateral line and olfactory organ) in fishes .	[U] / [CO3]
22.	Describe various types of growth in finfishes and explain their importance in fisheries biology .	[U] / [CO5]

[10x1 = 10]